The students decide not to use the bridge rectifier in the circuit and return to the original design of Figure 1. They next test the smoothing operation with a modified circuit shown in Figure 5 using C = 2000 μF, a switch, S, and a 100 Ω resistor. The switch is initially **open** so the resistor is disconnected at this time.

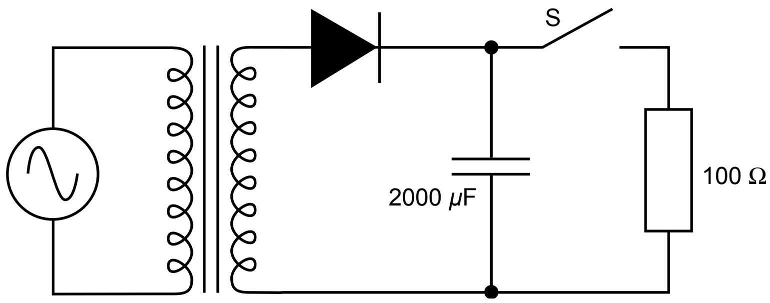

Figure 5

Question 5

The time constant of the circuit shown in Figure 5 is measured by the students to be 2 ms.

What is the effective forward resistance of the diode?

A. 0.1 Ω

B. 1 Ω

C. 2 Ω

D. 10 Ω

2 marks

The students now **close** the switch of the circuit of Figure 5, connecting the 100 Ω resistor into the circuit. The voltage across the 100 Ω resistor is measured using a CRO and is observed to comprise a DC component of about 12 V and an AC component, or ripple. They investigate this ripple by varying the values of some of the components in the circuit of Figure 5.

Question 6

Which one of the following statements (**A.–D.**) concerning the magnitude of the ripple voltage is correct?

A. It approximately doubles when the capacitance is halved to 1000 μF and the resistance doubled to 200 Ω.

B. It approximately doubles when the capacitance is doubled to 4000 μF and the resistance halved to 50 Ω.

C. It approximately doubles when the capacitance is halved to 1000 μF and the resistance left unaltered at 100 Ω.

D. It approximately doubles when the capacitance is doubled to 4000 μF and the resistance left unaltered at 100 Ω.

2 marks

The students now consider the final part of the design, the voltage regulator, using a resistor, R_V, in series with a 9 V Zener diode (see Figure 1).

The current-voltage characteristic of the Zener diode is shown in Figure 6.

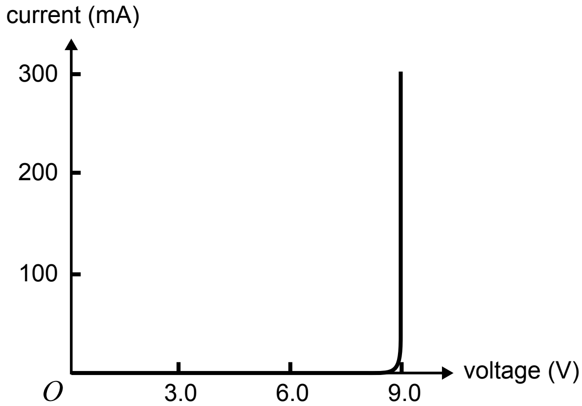

Figure 6

The students use a multimeter to measure the DC voltage across the 30 Ω resistor in Figure 1 to be approximately 3 V.

Question 7

Given that the voltage across the transformer is 24 V (peak-to-peak), in the space below explain why this value of 3 V across R_V is to be expected.

2 marks

Question 8

In this measurement for Question 7 the students could have used either a battery-powered multimeter (MM) or the mains-powered CRO.

Which one of the following statements concerning these two instruments is true?

A. The MM (in DC voltage mode) is the most sensible choice to measure this voltage across R_V compared to a CRO as it is less likely to affect the power supply circuit.

B. The CRO is the better voltage measuring instrument in this situation, as it can measure both the AC and DC voltages across R_V, while a MM can only measure DC voltages.

C. The MM is the better instrument to use as the only way to measure a current through a resistor is by using a MM (in AC or DC current mode). When using a CRO there is no way that the current in the resistor can be determined.

D. The CRO is the better instrument as it can be used to fully describe both the AC and DC voltages, while the MM cannot.

2 marks

Question 9

What is the power dissipated in the 30 Ω resistor, R_V, of Figure 1? Express your answer in mW.

mW

2 marks

The students now have a fully operational regulated 9 V DC power supply as shown in Figure 1. The final circuit has C = 2000 μF, a 9 V Zener diode regulator and R_V = 30 Ω. To start with, the students connect a load resistor of R_L = 300 Ω. They then do a series of tests to observe what might change the output DC across the load resistor.

Question 10

Based on their observations, which one of the following statements (**A.–D.**) is correct?

A. The output DC voltage changes if the mains supply voltage is increased to 250 V (RMS).

B. The output DC voltage changes if the load resistor is increased to 400 Ω.

C. The output DC voltage changes if the 9 V Zener diode is replaced by a 6 V Zener diode.

D. The output DC voltage changes if the capacitor is increased to 3000 μF.

2 marks

Question 11

The students have their circuit checked by a qualified electrician who advises them that their power supply circuit will require a heat sink. Explain why a heat sink is used in this regulated power supply circuit, and what part of the circuit needs to be attached to this heat sink.

2 marks

As a final part of their project to design and construct a regulated DC power supply, the students have to write a brief report. Students discuss safety and risk associated with this project.

Question 12

Which is the most important activity to ensure maximum safety and minimum risk?

A. learning from trial and error when designing and constructing electrical circuits

B. getting the teacher (or someone appropriately qualified) to check each stage of the design and construction

C. understanding how to correctly use a cathode ray oscilloscope and a multimeter for troubleshooting and diagnosis

D. understanding the physics of specific components such as the transformer, diodes, capacitor and resistor

2 marks

END OF QUESTION AND ANSWER BOOK